

Lab 3 - Architecture Justification

Name: Dhanush S SRN: PES1UG24AM360 Section: F

Scenario: Podcast Guest Scheduling & Outline Builder

Architecture Selection

We chose Layered Architecture for the Podcast Guest Scheduling & Outline Builder System.

Architectural Choice

The system is divided into four layers: Presentation (Guest Portal, Host Dashboard), Security (Auth Service), Business (Booking Manager, Outline Manager, Run-of-Show Generator) and Data (Scheduling Database). Each layer only communicates with the layer below it.

Reason 1

The system is small and has only two users, the guest and the host. Every feature follows the same flow: user interface, then business logic, then database. A layered design fits this flow, so the system is easy to build, understand and test.

Reason 2

A booking, the guest's talking points and the run-of-show sheet all belong to the same episode. Keeping them in one database keeps the data consistent. With microservices this data would be split across different services and would be harder to keep in sync.

Security Advantage

All requests from the Guest Portal and Host Dashboard go through the Auth Service first. It checks the login and the user's role, so only the host and the invited guest can view an episode. The business layer cannot be accessed directly.

Performance Benefit

The business and data layers run together, so the Run-of-Show Generator gets the approved topics using a direct function call and one database query instead of network calls between services. This helps generate the PDF production sheet in under 1 second.